GRAPE target spec: analytic local U_tgt vs full 2^n matrix

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Verdict. Simple analytic targets are enough. Standard GRAPE takes a small U_tgt (e.g. exp(-iθ ZZ), exp(-iα ZZZ/4), SNAP, or projector gadget I+(e^{-iγ}-1)C_c) on a few levels/modes — **not** the densified full-instance 2^n×2^n cost unitary. Instance U_c = exp(-iγ H_c) is a classical schedule of many frozen pulses.

What GRAPE optimizes

Khaneja et al. (2005) maximize fidelity between realized propagator U_f and a chosen target U_tgt (unitary synthesis) or a set of state transfers. U_tgt may be an analytic generator, a small explicit matrix, or a subspace map. Full densified n-qubit U is possible in principle but exponentially hard — not the literature default.

(a) Library GRAPE of local gadgets vs (b) full-instance GRAPE

(a) Compile each frozen clause gadget / ZZ / SNAP once on its small support → pulse library; classically schedule for the instance. No 2^n densify. Same physics as Arm C densified path.

(b) GRAPE the full instance U_c (densified E). Exp-scaling; Rowland & Jones note complexity scales badly with qubit number. Not our claim.

Citations

1. Khaneja et al., J. Magn. Reson. 172, 296 (2005). Target U_zzz(α)=exp{-iα I1z I2z I3z/4}. DOI 10.1016/j.jmr.2004.11.004 ; PDF tum.de ; arXiv quant-ph/0502173

2. Rowland & Jones, arXiv:1203.6260 — fidelity to target propagator; scales badly with #qubits.

3. Heeres et al., Nat. Commun. 8, 94 (2017), arXiv:1608.02430 — GRAPE logical/cavity gates on truncated space.

4. Fösel et al., arXiv:2004.14256 — target need not be defined on entire Hilbert space; logical subspace L OK.

5. Glaser et al., Eur. Phys. J. D 69, 279 (2015), arXiv:1508.00442 — QOC/GRAPE review.

6. Wilhelm et al., arXiv:2003.10132 — GRAPE intro for quantum technologies.

Plain answer for Zach

For exp(-iθ ZZ) or a projector gadget, GRAPE uses that small analytic U_tgt. You do not hand GRAPE the full 2^n instance U_c. Instance U_c is the product of many local pulses (classical schedule). Non-exp vs Arm C densified E; same U=exp(-iγ H_c) physics.

`Source: explore_OCT29/docs/GRAPE_TARGET_SPEC.md`